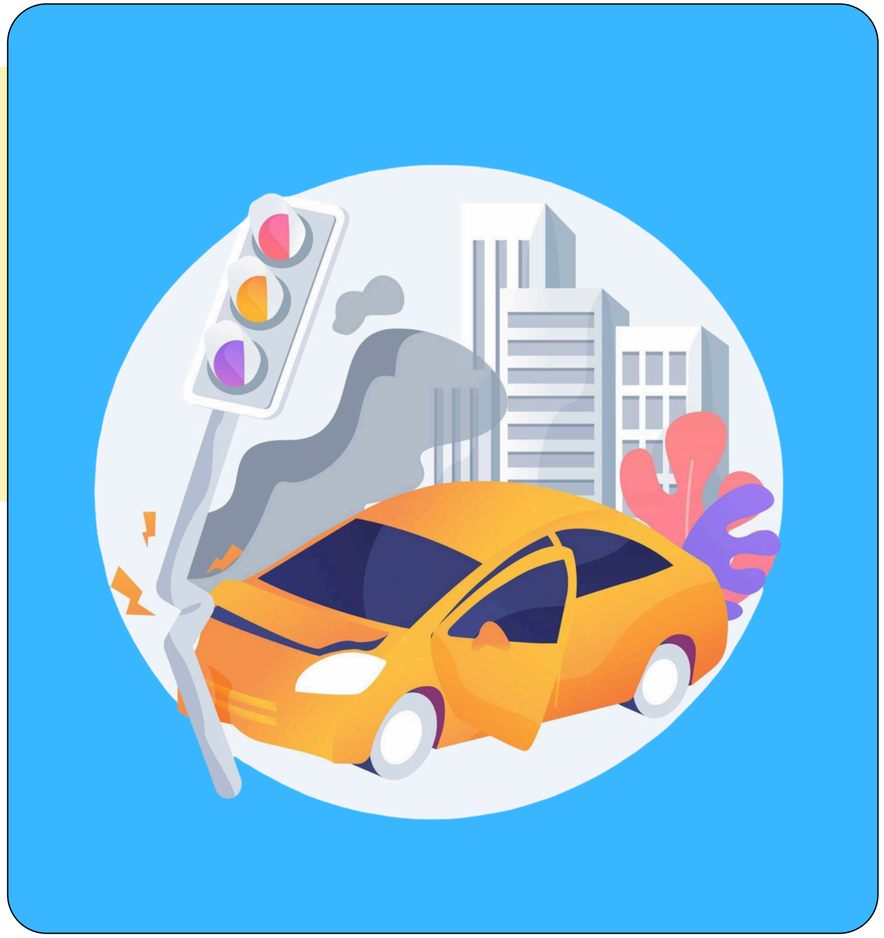
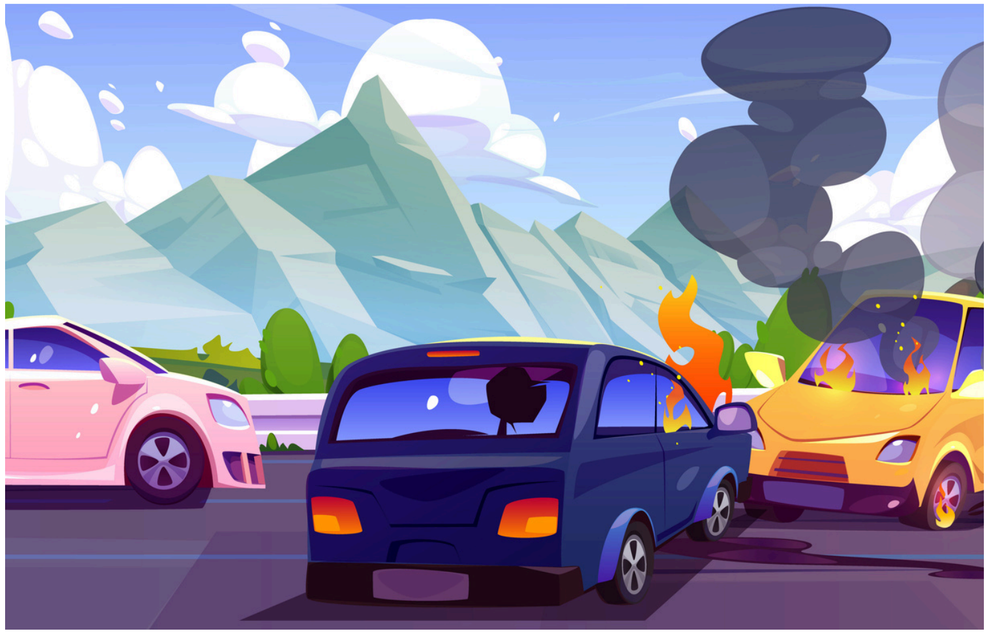


Real-Time Driver and Road Monitoring System for Risk Prediction

SLIIT
FACULTY OF COMPUTING

Main Objective

To develop a real-time AI and IoT-based driver monitoring system that integrates driver fatigue detection, driver drowsiness detection, road and weather condition monitoring, and personalized driver behavioural deviation detection to improve early risk prediction, driver awareness, and overall road safety.



Video Demonstration

Demonstration Video Link

Sub Objectives

01 Fatigue Detection For System for Long-Term Driver Monitoring

02 Multi-sensor Driver Drowsiness Detection using Physiological and Environmental Factors

03 Road-sign and weather-condition monitoring with in-vehicle real-time alerts.

04 Driver Behavioural Deviation Detection Based on Steering-Wheel Patterns

Problem

- How can real-time monitoring of driver condition, road environment, and personalized driving behavior be integrated to predict risks and reduce accidents?

- This study aims to design a multi-sensor, machine learning–based system that proactively enhances road safety through timely detection and alerts.

Overall Project

- Develop an ML and IOT based system for real time risk prediction
- Multi-sensor to Analyze Driver and Road Conditions
- Intergrate Driver and Road related data for conversation
- Aims to ensure the saftey of both pedistrians and drivers
- Involes multi component collabaration for slove Road saftey

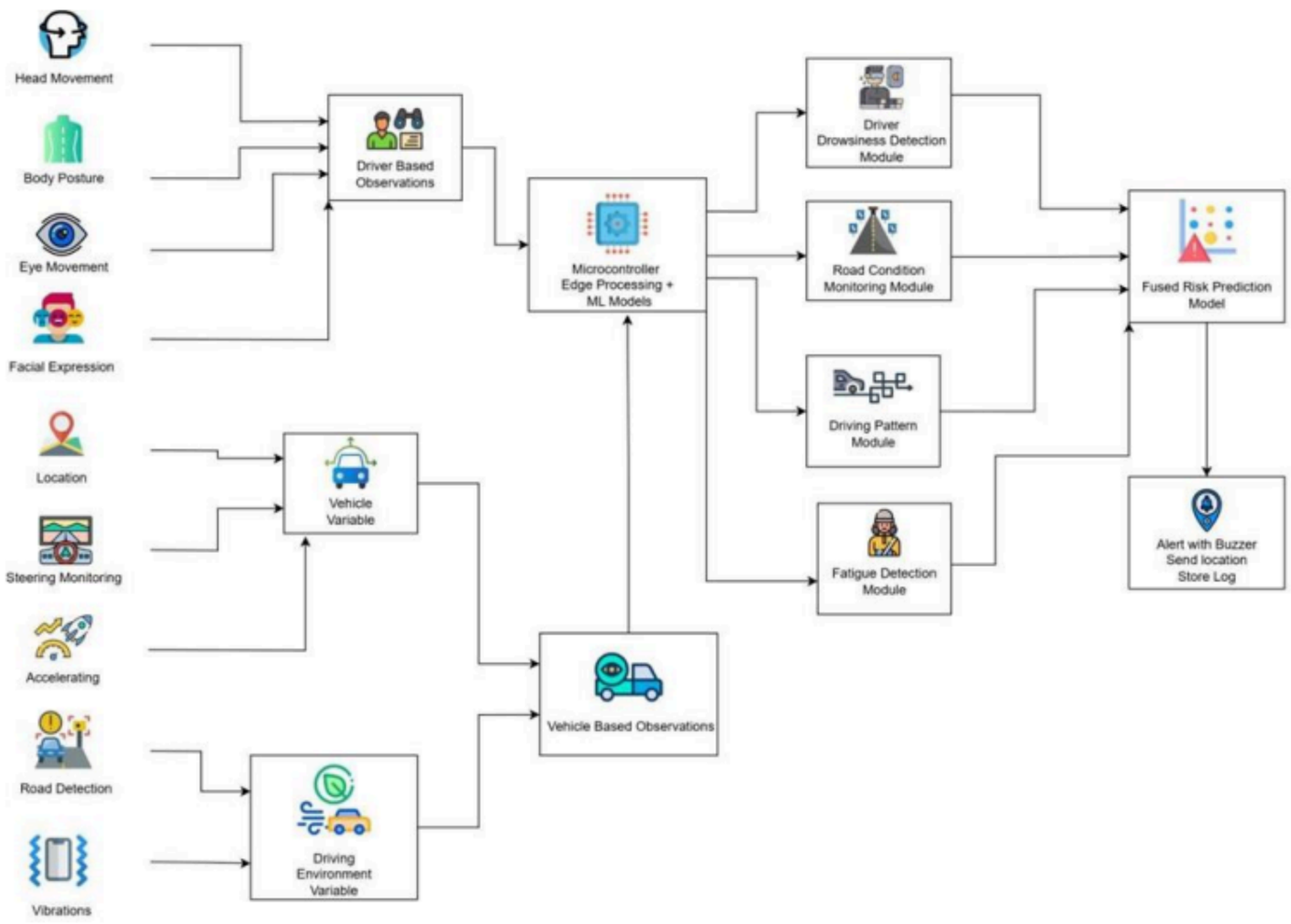
RESEARCH GAP

Current driver monitoring systems mainly rely on single detection methods, such as camera-based drowsiness detection or basic vehicle data. These systems do not integrate physiological signals, driver behaviour analysis, road monitoring, and environmental factors into one real-time system.

Existing systems:

- Detect drowsiness and fatigue using only visual indicators
- Do not combine physiological data such as HRV or SpO_2
- Lack traffic sign, speed, and weather monitoring
- Ignore personalized driving behaviour patterns

System Overview diagram



Commercialization

Current Target Users

sri lanka long route bus drivers, transportation companies, logistics and fleet management services, road safety authorities, and smart vehicle technology providers.

Value Proposition

A unified AI-powered driver monitoring system that integrates fatigue detection, driver drowsiness detection, road and weather condition monitoring, and personalized driver behaviour analysis to provide real-time risk prediction and enhance road safety.

Adoption Plan / Next Steps

Pilot implementation in selected vehicles → Evaluate system accuracy and usability → Improve AI models and sensor integration → Expand deployment to fleet management systems and smart transportation solutions.

Thank you so much